

Introduction to Artificial Intelligence and Machine Learning

Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on creating systems that can perform tasks that normally require human intelligence. AI systems can process information, recognize patterns, understand language, and support decision making.

Machine Learning

Machine Learning (ML) is a branch of AI in which algorithms learn patterns from data. Instead of explicitly programming every rule, a machine learning model uses examples to learn a relationship between input features and expected outputs.

Supervised Learning

Supervised learning uses labeled training data. Classification predicts categories such as spam or not spam, while regression predicts continuous values such as house prices.

Embeddings

Text embeddings convert text into numerical vectors. Similar pieces of text tend to have vectors that are close together in the embedding space. Embeddings are useful for semantic search and retrieval systems.

Vector Databases

Vector databases store numerical embeddings and make it possible to efficiently search for vectors that are similar to a query. ChromaDB is a vector database that can be used to store documents, embeddings, and metadata.

Retrieval Augmented Generation

Retrieval Augmented Generation (RAG) combines information retrieval with a large language model. Relevant document chunks are retrieved from a vector database and provided to the language model as context. The model then generates an answer based on the retrieved information.

Ollama

Ollama allows large language models to run locally. In this project, Ollama is used both for generating text embeddings and for generating an answer from retrieved document context.